The Anatomy of Album Sales \ Modeling the Diffusion of Everclear's *Sparkle and Fade* Album Sales

MKTG/STAT 4760/7760 | Project #2

March - April 2026

Abstract. Released in May 1995, Everclear's *Sparkle and Fade* followed a proposed three-phase commercial path: an initial plateau, a mid-cycle trough, and a Christmas breakthrough that permanently shifted the weekly sales level. Radio airplay explains the broad trend but not the discontinuity at week 31. The preferred **Burr XII + 2-Segment** model ($k = 14$, $\ell = -2{,}001{,}266$, BIC = 4,002,735, MAPE = 9.7%) seems to provide an accurate characterization of the data, a coherent behavioral story, and the comparatively strongest validation evidence out of the models in this paper. The implied March 1996 forecast is approximately **125,000 units** across the four withheld weeks. Multicollinearity among the airplay variables and sensitivity to optimization starting points affect parameter estimates in several specifications; this paper places primary emphasis on the generative story of each model.

1 Data

Sparkle and Fade shipped May 28, 1995. We observe 41 weekly unit sales and radio airplay observations through March 3, 1996 ($T = 41$, $N = 2{,}000{,}000$); four weeks in March 1996 act as the prediction period.

Table 1: Dataset summary: Everclear *Sparkle and Fade*.

Attribute	Value
Observation window	May 28, 1995 — March 3, 1996
Training weeks (T)	41
Holdout weeks	4 (March 10 — March 31, 1996)
Market size (N)	2,000,000
Total in-sample sales	339,405 (16.97% penetration)
Peak weekly sales	31,929 (week 39, Feb. 17, 1996)
Mid-cycle trough	Wks 15–20: airplay nadir; sales fall to 1,726
Breakthrough (wk 31)	Dec. 24, 1995: sales surge 9,832 → 17,642 (+79%)

Three structural features define the empirical target for every modeling choice in this paper. (1) Weeks 1–14 show moderate launch sales despite strong early airplay: radio exposure must *accumulate* before adoption scales, indicating a stock-of-awareness mechanism rather than a pure contemporaneous flow effect. (2) Weeks 15–20 trace a clear trough as airplay collapses to its sample minimum, confirming that both directions of the radio–sales co-movement are real

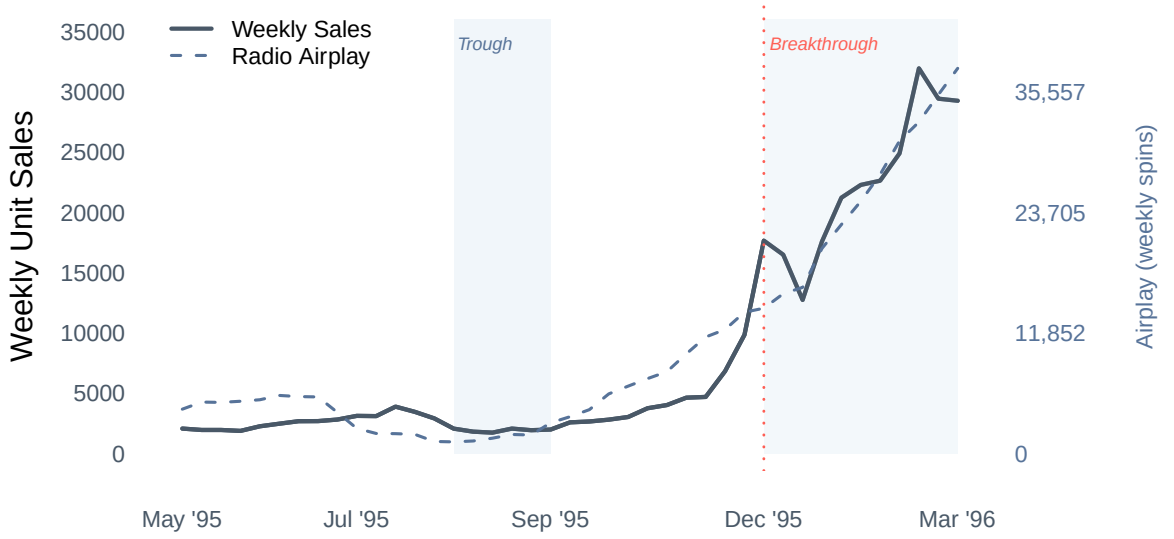


Figure 1: Weekly sales (dark) and radio airplay (blue dashed, right axis). Shaded regions mark the mid-cycle trough (blue) and post-breakthrough phase (light blue). The dotted vertical marks the Christmas breakthrough (wk 31).

and consequential. (3) The week 31 surge (+79%) coincides with Christmas, escalating MTV rotation, and chart crossover into mainstream territory—not an airplay spike. Any credible model must reproduce all three regularities.

A critical external anchor for the data: lead single **"Santa Monica" was formally added to US rock radio playlists on September 19, 1995** - week 17 of the sales series. Prior to this national add, airplay was somewhat limited to regional and alternative-format stations, which explains the low and declining airplay trajectory through weeks 1–16 and the corresponding sales trough. Week 17 marks the onset of sustained national rotation, after which cumulative airplay began accumulating rapidly. By week 30, the stock of prior exposure had reached levels sufficient to prime the genre-committed segment for adoption; the week 31 Christmas/chart breakthrough then triggered the second, mainstream wave. This external timing corroborates the two-segment architecture: the cumulative airplay stock variable directly encodes the "Santa Monica'' airplay accumulation curve that drove Segment 1, while the breakthrough dummy captures the particular point in time that unlocked Segment 2.

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2 Modeling Framework

We adopt the **Timing of First Purchase (TFP)** framework: each of $N = 2{,}000{,}000$ potential consumers has a latent adoption week drawn from a continuous distribution. The aggregate log-likelihood over $T = 41$ training weeks is:

$$\ell(\boldsymbol{\theta}) = \sum_{t=1}^{T} n_t \ln p_t + \left(N - \textstyle\sum_t n_t\right) \ln[1 - F(T)],$$

where $p_t = F(t) - F(t-1)$. The second term is critical at $N = 2{,}000{,}000$: it accounts for the 1,660,595 consumers who had not yet adopted by week 41, treating them as right-censored at the end of the training window. Ignoring this term would inflate estimated adoption rates by assigning zero probability to non-adoption, biasing every parameter upward.

Time-varying covariates enter through a power-law cumulative exposure function. All covariates are standardized (mean 0, unit variance) before entry, so each coefficient $\hat{\beta}_j$ measures the log-hazard response to a one-SD shift and multipliers $e^{\hat{\beta}_j}$ are directly comparable across models and segments. Four distributional families that we will explore are defined below.

Exponential *(constant hazard; no duration dependence)*
Let $A(t) = \sum_{k=1}^{t} \exp(\boldsymbol{\beta}^\top \mathbf{z}_k)$ (cumulative covariate exposure). Then:

$$F(t) = 1 - e^{-\lambda A(t)}, \qquad h(t) = \lambda \exp(\boldsymbol{\beta}^\top \mathbf{z}_t).$$

The hazard is constant in t; all temporal dynamics must be absorbed by covariates. Heterogeneity across consumers is not modelled.

Weibull *(flexible power-law hazard; no Gamma mixing)*
Let $B(t) = \sum_{k=1}^{t} [k^c - (k-1)^c] \exp(\boldsymbol{\beta}^\top \mathbf{z}_k)$. Then:

$$F(t) = 1 - e^{-\lambda B(t)}, \qquad h(t) = \lambda c\, t^{c-1} \exp(\boldsymbol{\beta}^\top \mathbf{z}_t).$$

$c > 1$ gives an increasing hazard (imitators); $c < 1$ a declining one (innovators dying off). Without Gamma mixing, all consumer heterogeneity must be captured through covariates.

Pareto II (Lomax) *(declining hazard; Gamma heterogeneity)*
Using the same $A(t)$ as the Exponential:

$$F(t) = 1 - \left(\tfrac{\alpha}{\alpha + A(t)}\right)^r, \qquad h(t) = \frac{r \exp(\boldsymbol{\beta}^\top \mathbf{z}_t)}{\alpha + A(t)}.$$

Arises from Gamma(r, α) mixing of an Exponential hazard. As fast adopters exit the risk set, slower adopters accumulate disproportionately among survivors, driving the population hazard down. Smaller r implies greater unobserved heterogeneity.

Burr XII *(flexible hazard; Gamma heterogeneity + power-law clock)*
Using the same $B(t)$ as the Weibull:

$$F(t) = 1 - \Big(\frac{\alpha}{\alpha + B(t)}\Big)^r, \qquad h(t) = \frac{r\, c\, t^{c-1} \exp(\boldsymbol{\beta}^\top \mathbf{z}_t)}{\alpha + B(t)}.$$

Nests the Pareto II at $c = 1$ (since $B(t) \to A(t)$ and $c\,t^{c-1} \to 1$). The shape parameter c introduces duration dependence through both the transformed clock $B(t)$ and the $c\,t^{c-1}$ numerator term, allowing unimodal or declining hazard profiles. Gamma heterogeneity is preserved alongside this flexibility, making the Burr XII the most structurally rich single-segment family considered.

Fit metrics. BIC $= -2\ell + k\ln(N)$, N = 2,000,000; MAPE $= \frac{1}{41}\sum_t |n_t - \hat{n}_t|/n_t$; LLRT $= 2(\ell_{\text{complex}} - \ell_{\text{simple}}) \sim \chi^2_{\Delta k}$.

When single-segment residuals show systematic structure, we extend to a **two-segment mixture** $p(t) = \pi\, p_1(t;\boldsymbol{\theta}_1) + (1-\pi)\, p_2(t;\boldsymbol{\theta}_2)$, with π the mixing weight and each segment carrying its own family, parameters, and covariate vector.

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3 Act I: Baseline Models

Table 2: Baseline (no-covariate) models.

Model	k	r	α	c	Log-Likelihood	BIC	MAPE
Pareto II	2	8,143,769	1,887,396,628	—	−2,187,695	4,375,419	170.2%
Burr XII	3	917.7	107,539,461	2.685	−2,063,438	4,126,920	87.8%

Without covariates, the Pareto II collapses to a near-flat hazard ($r \approx 8 \times 10^6$, nearly homogeneous), while the Burr XII uses $c = 2.685$ to produce a smooth unimodal arc. Both are structurally blind to market dynamics: they establish a necessary empirical lower bound and demonstrate unambiguously that covariates are essential, but they contribute nothing to the behavioral story.

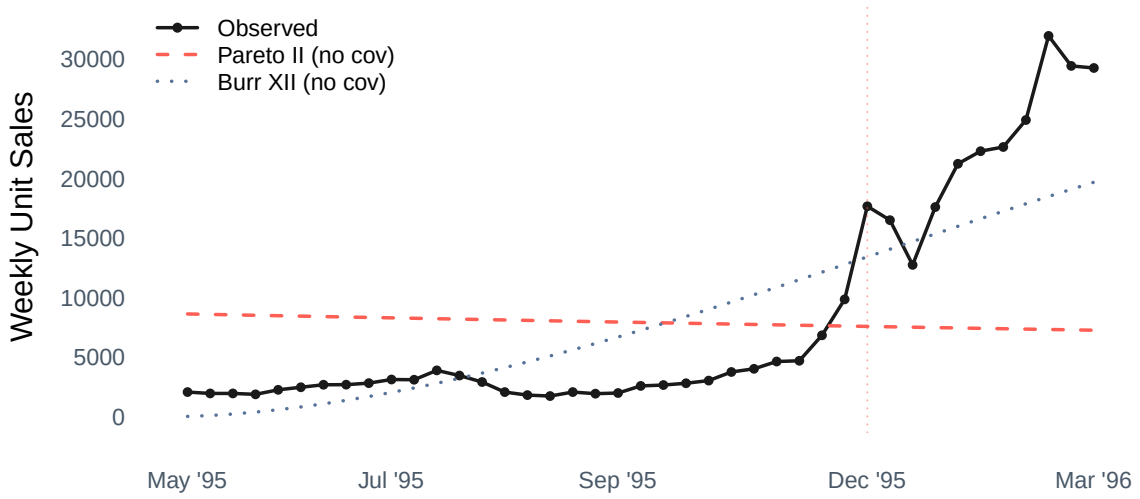


Figure 2: Baseline models vs. observed sales. Pareto II (coral dashed): near-flat hazard. Burr XII (blue dotted): smooth unimodal arc. Neither reproduces the trough, breakthrough, or any radio-correlated feature.

Parsimony: Excellent ($k = 2$–3). Required as a benchmark; no covariate-driven complexity.

Performance: Disqualifying. MAPE = 170.2% (Pareto II) and 87.8% (Burr XII); structurally blind to all three data regularities.

Robustness: Not applicable; systematic bias makes holdout testing meaningless. But, the parameter estimates are robust and do not vary based on optimization starting point.

Generative story: None. Calendar time alone cannot explain the trough, breakthrough, or radio dynamics. *Verdict: necessary test to understand that covariates are needed.*

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4 Act II: Radio Airplay

Radio is the album's primary promotional lever. Introducing a single standardized airplay covariate immediately unlocks $\Delta\text{BIC} > 96{,}000$ over the no-covariate Burr XII—decisive evidence that the radio–sales linkage is real and economically substantial.

Table 3: Single-covariate airplay models. β_{AP} is standardized.

Model	k	$\hat{\beta}_{\text{AP}}$	Log-Likelihood	BIC	MAPE
Exp + Airplay	2	0.820	−2,021,612	4,043,254	38.3%
Pareto II + Airplay	3	1.017	−2,021,762	4,043,567	38.6%
Burr XII + Airplay	4	0.838	−2,015,104	4,030,266	32.2%

The Burr XII + Airplay leads the tier ($\Delta\ell = 6{,}508$ over Exp), with a one-SD airplay shift multi-

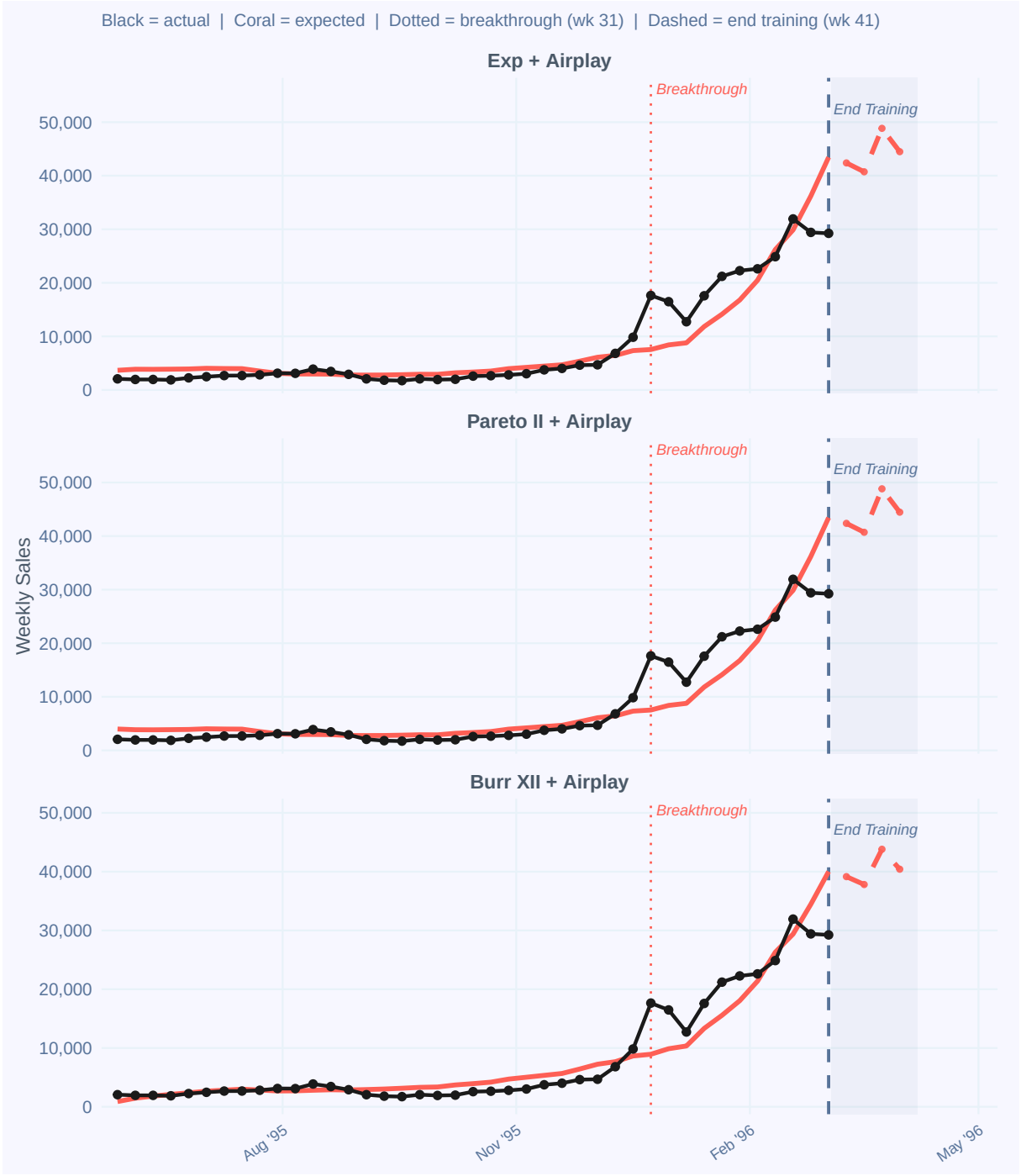


Figure 3: Airplay-only fits. All three track radio-correlated phases but miss the week 31 breakthrough and produce exploding holdout forecasts.

plying the hazard by $e^{0.838} \approx 2.3\times$. Yet two structural failures persist. First, the breakthrough is chronically underpredicted (forecast approx. 8,900 vs.
actual 17,642 at week 31) because airplay that week was not unusually high—the model lacks any mechanism to generate the discontinuous demand surge. Second, holdout forecasts explode past 40,000 by extrapolating the rising airplay trend into unconstrained territory. These are not estimation failures—they expose a structurally incomplete behavioral theory: contemporaneous radio flow alone cannot generate threshold shifts or self-limiting post-breakthrough dynamics.

Parsimony: Good ($k = 2$–4); each addition justified by the improvement in LL and LLRT where applicable. ΔBIC $> 96{,}000$ over baseline.

Performance: Moderate. Burr XII + Airplay MAPE = 32.2%; systematic breakthrough underprediction.

Robustness: Weak. Holdout forecasts explode; model extrapolates airplay trend without a depletion mechanism.

Generative story: Establishes the radio–sales linkage but misses the breakthrough mechanism entirely. *Verdict: necessary stepping stone; structurally incomplete.*

5 Act III: A Richer Covariate Set

The Act II failures share a root cause: the model is missing two distinct channels of demand activation. **Lagged (1 week) airplay** (replacing contemporaneous) is predetermined at the moment of purchase, eliminating endogeneity and improving forecast stability. A **breakthrough dummy** ($\mathbf{1}_{t\geq 31}$) directly absorbs the permanent regime shift at week 31. **Cumulative airplay** (standardized) captures the stock of prior radio exposure that drives the pre-breakthrough ramp.

Table 4: Covariate screening decisions.

Covariate	Status	Rationale
Lagged airplay (Air_{t-1})	Retained	Determined the week before current sales; no endogeneity from radio responding to popularity
Breakthrough dummy ($\mathbf{1}_{t\geq 31}$)	Retained	Absorbs permanent structural shift at wk 31; multiplicative hazard uplift
Cumulative airplay	Retained (2-Seg only)	Captures stock of prior exposure; key for genre-fan segment
Holiday dummy	Retained (Weibull)	Identifies holiday gift-buying; significant in Weibull specification
Contemporaneous airplay	Dropped	Endogeneity risk; dominated by lagged level
Change in airplay	Dropped	Near-redundant given lagged level; endogeneity risk

Table 5: CleanCov single-segment models. ΔBIC $> 15{,}000$ over airplay tier (Weibull best: ΔBIC = 20,992). Pareto II uses $A(t)$ clock (c not applicable).

Model	k	c	$\hat{\beta}_{\text{lag}}$	$\hat{\beta}_{\text{hol}}$	$\hat{\beta}_{\text{brk}}$	Log-Likelihood	BIC	MAPE
Pareto II + CleanCov	4	—	0.415	—	1.216	−2,007,385	4,014,827	24.0%
Burr XII + CleanCov	5	1.256	0.382	—	1.026	−2,005,174	4,010,420	19.4%
Weibull + CleanCov	5	1.231	0.427	0.174	0.941	−2,004,601	4,009,275	18.0%

The Weibull and Burr XII CleanCov models achieve ΔBIC $> 19{,}000$ over the airplay tier with MAPE 18–19%; Pareto II + CleanCov ($k = 4$, no time trend) achieves ΔBIC = 15,439 and MAPE = 24%. Because all covariates are standardized to mean 0 and unit variance, each $\hat{\beta}_j$ is

directly interpretable as the log-hazard response to a one-SD increase, and the hazard multiplier $e^{\hat{\beta}_j}$ is comparable across models and segments. The breakthrough coefficient is economically dominant in every family: $e^{1.026} \approx 2.8\times$ (Burr XII), $e^{0.941} \approx 2.6\times$ (Weibull), and $e^{1.216} \approx 3.4\times$ (Pareto II), all confirming that week 31 opens a new demand regime rather than being a high-airplay week. Lagged airplay multipliers are strikingly consistent at $e^{0.38\text{–}0.43} \approx 1.5\times$ per SD across all three families, confirming a robust and endogeneity-free radio–sales link. The Weibull holiday coefficient $e^{0.174} \approx 1.2\times$ captures the Christmas gift-buying uplift—a real but modest effect relative to the breakthrough itself ($\approx 2.6\times$). The Burr XII shape parameter $c = 1.256 > 1$ indicates a mildly increasing hazard profile, reflecting rising consumer awareness as cumulative airplay builds; Weibull shows $c = 1.231$, consistent with the same pattern.

Table 6: Asymptotic inference for CleanCov covariate and shape parameters (rows grouped by model). SE from numerical Hessian; $t = \hat{\theta}/\text{SE}$, $H_0 : \theta = 0$. All entries $p \ll 0.001$ ($***$). $e^{\hat{\beta}}$ = per-SD hazard multiplier; "—" for shape c.

Param	$\hat{\theta}$	$e^{\hat{\theta}}$	SE	t	Sig
Pareto II + CleanCov					
$\hat{\beta}_{\text{lag}}$	0.415	1.51×	0.00249	166.7	***
$\hat{\beta}_{\text{brk}}$	1.216	3.37×	0.00659	184.5	***
Burr XII + CleanCov					
c	1.256	—	0.00414	303.4	***
$\hat{\beta}_{\text{lag}}$	0.382	1.46×	0.00237	161.2	***
$\hat{\beta}_{\text{brk}}$	1.026	2.79×	0.00704	145.7	***
Weibull + CleanCov					
c	1.231	—	0.00419	293.8	***
$\hat{\beta}_{\text{lag}}$	0.427	1.53×	0.00298	143.3	***
$\hat{\beta}_{\text{hol}}$	0.174	1.19×	0.00504	34.5	***
$\hat{\beta}_{\text{brk}}$	0.941	2.56×	0.00728	129.3	***

All covariate parameters are statistically decisive: the smallest t-statistic across any model–covariate pair is $t = 34.5$ (Weibull holiday, $p < 10^{-260}$), confirming that every retained covariate contributes real explanatory power and is not a product of overfitting. The striking cross-model consistency of $\hat{\beta}_{\text{lag}} \approx 0.38$–$0.43$ (hazard multiplier $\approx 1.5\times$/SD) and $\hat{\beta}_{\text{brk}} \approx 0.94$–$1.22$ (multiplier 2.6–3.4×) confirms that the radio–sales mechanism and the breakthrough regime shift are genuine structural features of the data, robust to distributional family choice. The shape parameters $c > 1$ in both Burr XII and Weibull confirm mildly increasing base hazard before the breakthrough—consistent with growing consumer awareness driven by the "Santa Monica'' airplay accumulation. The holiday effect, present only in Weibull ($t = 34.5$, $e^{0.174} \approx 1.2\times$), reflects a modest Christmas gift-buying uplift that the Burr XII's Gamma heterogeneity absorbs into its distributional structure rather than requiring a separate coefficient.

However, the residuals are systematic. Weeks 32–41 are persistently over-predicted (+4.7% to +66.4% in holdout), and weeks 28–30 are under-predicted. The underlying cause is structural: the same breakthrough coefficient that correctly calibrates week 31 keeps too large a survivor pool active in subsequent weeks. No re-estimation or additional covariate can fix this within a single segment, because the problem is not parameter error—it is model misspecification.

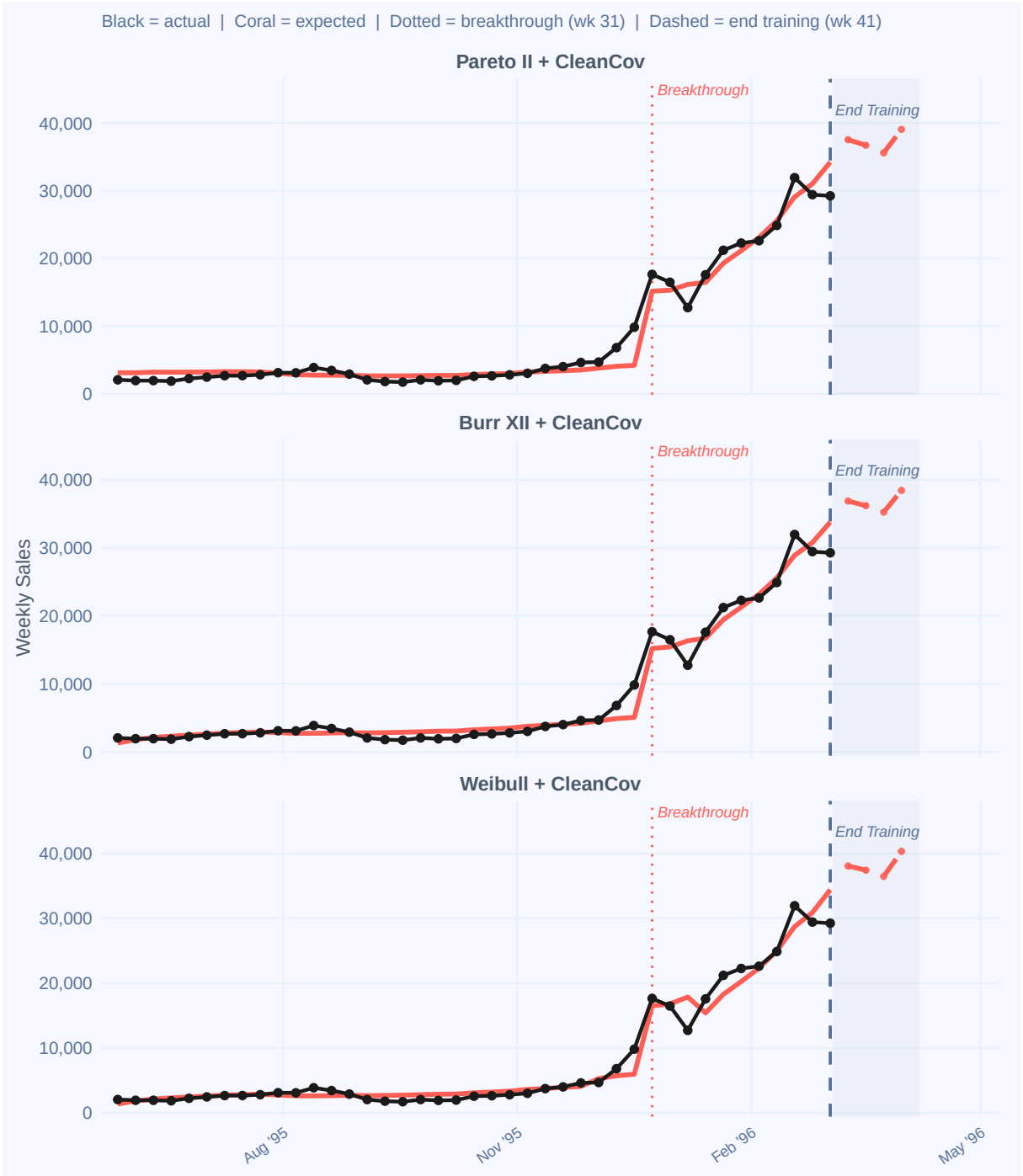


Figure 4: CleanCov fits. All three reproduce the breakthrough spike, but over-prediction in weeks 32–41 signals that a single segment cannot simultaneously calibrate the breakthrough magnitude and the post-breakthrough depletion rate.

Table 7: Weibull + CleanCov holdout (estimated wks 1–35). Persistent upward error drift, reaching +67.4% by week 41.

Week	Date	Actual	Holdout Pred.	Holdout Err.	Full-Sample Pred.	Full-Sample Err.
36	Jan 28, 1996	22,262	22,199	-0.3%	20,242	-9.1%
37	Feb 04, 1996	22,612	25,707	13.7%	22,322	-1.3%
38	Feb 11, 1996	24,875	30,426	22.3%	24,979	0.4%
39	Feb 18, 1996	31,929	37,482	17.4%	28,718	-10.1%
40	Feb 25, 1996	29,413	41,753	42.0%	30,888	5.0%
41	Mar 03, 1996	29,239	48,937	67.4%	34,390	17.6%

Parsimony: Good ($k = 5$ each); ΔBIC $> 20{,}000$ over airplay tier easily justifies covariate additions.

Performance: Strong in-sample. Best BIC = 4,009,275; MAPE = 18.0%. Substantial improvement over airplay tier.

Robustness: Mixed. Weibull + CleanCov holdout MAPE = 27.2%, with persistent upward error drift reaching +67.4% by wk 41; structural over-prediction confirmed.

Generative story: Convincing covariate rationale. However, systematic post-breakthrough over-prediction reveals that a single behavioral segment cannot reconcile the breakthrough jump and the subsequent depletion rate. *Verdict: best single-segment specification; residual structure may benefit from a two-population extension.*

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6 Act IV: Two-Segment Mixture

6.1 Theoretical Motivation

The Act III misfits are difficult to attribute to noise alone; they are more plausibly a diagnostic signature of latent population heterogeneity in *how* consumers respond to market signals, not merely in their adoption rates.

Segment 1 — Genre-Committed Fans (8.4%). Pre-existing alternative and indie rock listeners who are already primed for the album's sound—they may simply not yet have encountered the specific release. When "Santa Monica" reaches their radio dial, the genre fit appears immediate and a purchase decision can follow quickly: $\hat{\beta}_{\text{lagAP}} = 15.05$ encodes a striking sensitivity to the *weekly flow* of radio signal among this small, genre-invested audience. The mildly increasing base hazard ($c = 1.268$) is consistent with accumulating awareness within this community as national airplay grows from week 17 onward. The large breakthrough premium ($e^{3.75} \approx 42\times$) is plausibly the final confirmation signal: Christmas chart success provides even the most hesitant genre fan with the social validation that this is a worthwhile purchase. The near-zero cumulative-airplay response suggests that this cohort reacts primarily to *recent* radio signals rather than long-run stock—consistent with genre-informed listeners who tend to recognise the song on first or early encounter rather than needing extended repeated exposure.

Segment 2 — Mainstream Chart Followers (91.6%). The dominant cohort, plausibly representing general music consumers—listeners who enjoy popular music broadly but for whom adoption of any specific album tends to require the gradual accumulation of awareness before action. For this group, the radio stock needs to build ($\hat{\beta}_{\text{cumAP}} = 3.75$, $e^{3.75} \approx 42\times$/SD), MTV rotation to escalate, and the album to appear on best-seller charts before it enters their active consideration set. The moderate breakthrough premium ($e^{0.544} \approx 1.7\times$) is consistent with the tipping-point signal of Christmas season legitimation; the near-constant hazard thereafter ($c = 0.926$) sustains steady purchasing as the album enters broader cultural circulation. A large consumer base that tends to convert after social proof and repeated exposure have lowered the perceived risk of an unfamiliar artist.

The two-segment log-likelihood is $\ell(\pi, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = \sum_t n_t \ln[\pi p_1(t) + (1-\pi)p_2(t)] + (N - \sum_t n_t) \ln[\pi S_1(T) + (1-\pi)S_2(T)]$.

6.2 Parameter Estimates

Table 8: Weibull + 2-Segment estimates ($k = 11$, LL = −2,004,611, BIC = 4,009,382). The 2-segment extension yields $\Delta\ell = -10$ versus single-segment Weibull + CleanCov (LL worsens); BIC *worsens*.

Parameter	Segment 1	Segment 2
λ	0.001384	0.000908
c	0.397	1.274
$\hat{\beta}_{\text{lagAP}}$	1.071	0.399
$\hat{\beta}_{\text{Holiday}}$	≈ 0	0.094
$\hat{\beta}_{\text{brk}}$	0.582	0.985
$\hat{\beta}_{\text{cumAP}}$	≈ 0	≈ 0
Mixing weight π / $(1-\pi)$	**17.9%**	**82.1%**

Why the Weibull + 2-Segment underperforms—and why this matters. The Weibull achieves a log-likelihood of −2,004,611 with two segments— fractionally *worse* than single-segment Weibull + CleanCov (−2,004,601), a net $\Delta\ell = -10$ from adding $\Delta k = 6$ parameters; BIC worsens accordingly. This outcome is unlikely to be a simple estimation failure. The Weibull hazard $h(t) = \lambda c\, t^{c-1} e^{\boldsymbol{\beta}^\top \mathbf{z}_t}$ incorporates no unobserved individual heterogeneity—all consumers

face the same hazard up to observed covariates. Without Gamma mixing, the optimizer struggles to sustain two populations with structurally different hazard trajectories, because there is no distributional mechanism to generate the required dispersion in individual adoption times. The Burr XII's Gamma mixing provides this latent heterogeneity, giving the optimizer room to carve out an increasing-hazard genre cohort ($c_1 = 1.268$) alongside a near-constant mainstream cohort ($c_2 = 0.926$). The Weibull shortfall is thus consistent with the interpretation that Gamma heterogeneity, rather than a richer covariate set, is the key structural enabler of the two-segment architecture.

Table 9: Burr XII + 2-Segment estimates ($k = 14$, LL = −2,001,266, BIC = 4,002,735). Seg. 1 = Genre-Committed Fans; Seg. 2 = Mainstream Chart Followers. † = non-significant / fixed at 0.

Parameter	Seg. 1 (8.4%)	Seg. 2 (91.6%)
r	0.00261	0.01606
α	58.29	0.292
c	1.268	0.926
$\hat{\beta}_{\text{lagAP}}$	15.05	$0^{\dagger}$
$\hat{\beta}_{\text{Holiday}}$	$0.010^{\dagger}$	—
$\hat{\beta}_{\text{brk}}$	3.750	0.544
$\hat{\beta}_{\text{cumAP}}$	0	3.750
Mixing weight π / $(1-\pi)$	**8.4%**	**91.6%**

Interpreting the Burr XII parameters through the two-segment lens. Segment 1 (genre-committed fans, 8.4%): the extreme lagged-airplay sensitivity ($\hat{\beta}_{\text{lagAP}} = 15.05$) reflects a small cohort of genre-invested listeners for whom hearing `Santa Monica` on radio is immediately sufficient to trigger purchase - a recognition-and-buy mechanism. Because these consumers already know they like this style of music, the radio encounter functions as a discovery event rather than a persuasion process, generating a near-instantaneous hazard spike. The increasing base hazard ($c = 1.268$) is consistent with this mechanism: it reflects not a slow individual decision process, but the expanding national reach of `Santa Monica` itself—as Capitol Records' radio promotion secured carriage on progressively more stations following the September 19 add, the weekly probability that any given genre fan *encountered* the song for the first time rose steadily through weeks 17–30, so the population-level adoption rate increases even though each individual's response upon hearing it remains immediate. The large breakthrough premium ($e^{3.750} \approx 42\times$) captures the final confirmation signal for any genre fans who were still on the fence by Christmas week. The zero cumulative-airplay response distinguishes them sharply from Segment 2: these consumers do not need months of stock accumulation—a single encounter with the right song is enough. Segment 2 (mainstream chart followers, 91.6%): adoption appears to be governed by the cumulative stock of radio exposure ($\hat{\beta}_{\text{cumAP}} = 3.750$, $e^{3.750} \approx 42\times$/SD), consistent with consumers who tend to require repeated radio encounters, chart visibility, and word-of-mouth before committing. The near-constant hazard ($c = 0.926$) sustains steady purchasing through the post-breakthrough period, and the moderate breakthrough uplift ($e^{0.544} \approx 1.7\times$) is interpretable as a Christmas-season social legitimation effect: the album's simultaneous appearance on bestseller racks during the peak gift-buying window—coinciding with `Santa Monica` reaching its national chart peak in late December 1995—created a self-reinforcing visibility cascade that lowered the perceived risk of purchase for consumers who had been accumulating awareness but had not yet committed. The near-zero estimates elsewhere add further interpretive coherence: $\hat{\beta}_{\text{lagAP}} \approx 0$ in Segment 2 suggests that chart followers respond primarily to accumulated rather than weekly radio exposure, while $\hat{\beta}_{\text{Holiday}} \approx 0$ in Segment 1 is consistent

with genre fans whose adoption is driven by musical recognition rather than the seasonal context.

6.3 In-Sample Fit and Holdout Validation

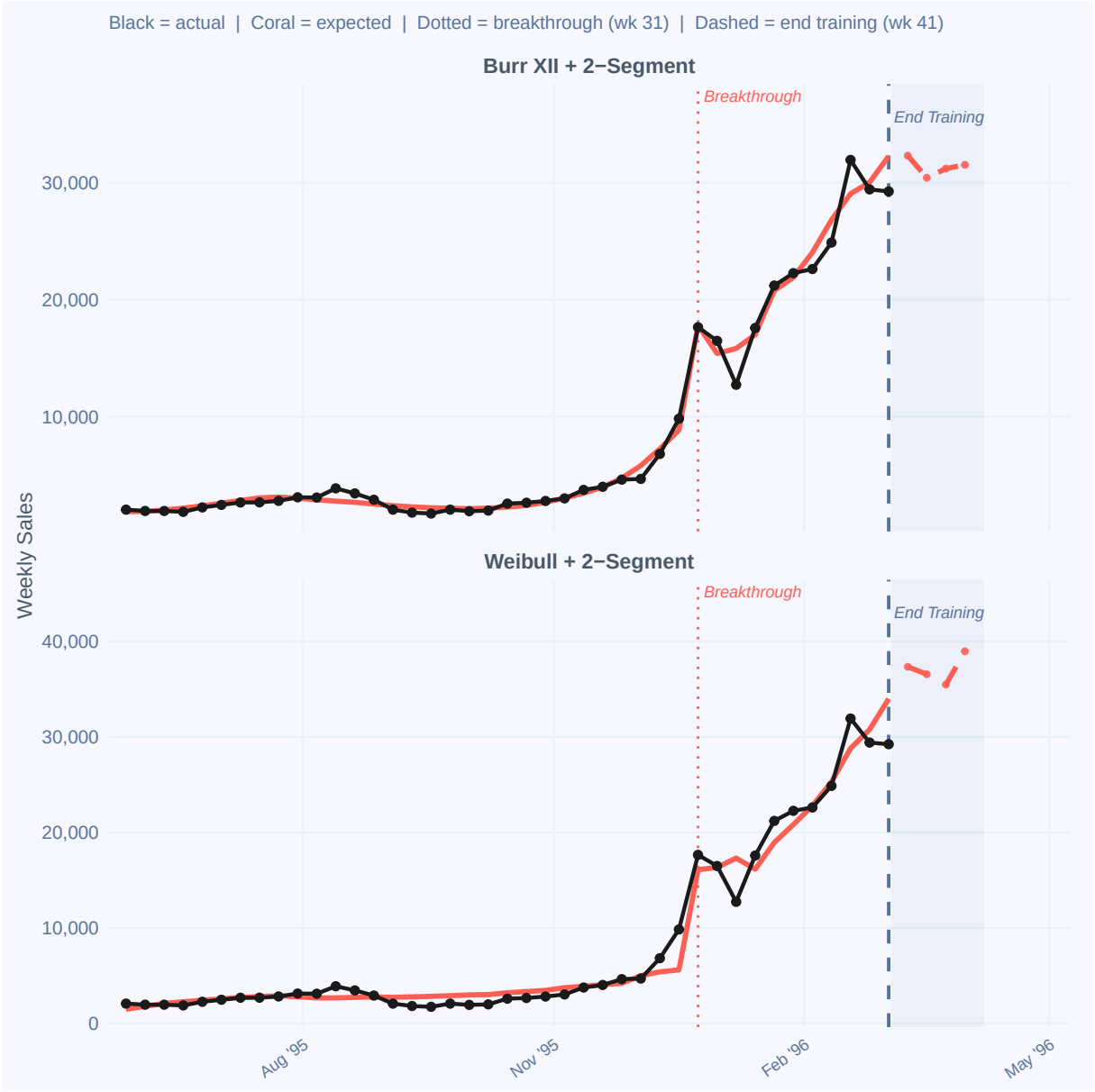


Figure 5: Two-segment fits. The Burr XII + 2-Segment achieves near-perfect fit at the breakthrough (17,806 vs. actual 17,642) and tracks post-breakthrough depletion. The Weibull + 2-Segment provides no improvement over its single-segment counterpart.

The Burr XII + 2-Segment substantially addresses all three Act III failure modes. At the breakthrough (week 31): 17,806 predicted vs. 17,642 actual—a 0.9% miss, compared with approximately 6–14% underprediction across the CleanCov tier (Weibull: −6.4%; Burr XII: −13.9%). Post-breakthrough (weeks 32–41): the systematic overprediction seen in single-segment models is largely eliminated as Segment 2 depletes naturally after the breakthrough event. On the pre-breakthrough ramp (week 30): 8,852 predicted vs. 9,832 actual (10.0% miss, versus approx-

Table 10: Point-by-point fit at key weeks.

Week	Actual	P2+FullCov	Burr XII+CleanCov	Burr XII+2-Seg	Weibull+2-Seg
1	2,061	1,980	1,303	1,896	1,470
10	3,113	2,356	2,839	3,033	2,772
20	1,980	3,112	3,069	2,189	3,005
29	6,819	5,292	4,896	7,254	5,388
30	9,832	5,733	5,074	8,852	5,592
31	**17,642**	**14,512**	**15,195**	**17,806**	**16,090**
32	16,476	15,406	15,426	15,393	16,332
38	24,875	26,403	25,552	26,805	25,293
40	29,413	30,397	30,744	30,016	30,780
41	29,239	32,808	33,749	32,253	34,001

imately 39–48% underprediction in the CleanCov tier), consistent with Segment 2's cumulative airplay stock gradually activating mainstream consumers in the weeks immediately preceding the social-legitimation tipping point.

Table 11: Burr XII + 2-Segment holdout (estimated wks 1–35; wks 36–41 are genuine out-of-sample). No systematic drift; holdout and full-model March forecasts agree within approximately 3%.

Week	Date	Actual	Holdout Pred.	Holdout Err.	Full-Sample Pred.	Full-Sample Err.
36	Jan 28, 1996	22,262	21,495	-3.4%	21,896	-1.6%
37	Feb 04, 1996	22,612	23,606	4.4%	24,025	6.2%
38	Feb 11, 1996	24,875	26,243	5.5%	26,805	7.8%
39	Feb 18, 1996	31,929	28,165	-11.8%	29,037	-9.1%
40	Feb 25, 1996	29,413	29,552	0.5%	30,016	2.1%
41	Mar 03, 1996	29,239	31,497	7.7%	32,253	10.3%

Table 12: Holdout comparison (wks 1–35 → 36–41). Burr XII + 2-Segment dominates on every metric.

Model	Training	MAPE	RMSE	Bias	Diagnostic
Burr XII + 2-Segment	**Weeks 1–35**	**5.6%**	**1,947**	**+38**	**No drift; forecasts stable**
Weibull + CleanCov	Weeks 1–35	27.2%	10,096	+7,696	Systematic over-prediction

The holdout results are compelling: MAPE = **5.6**% with no systematic drift (signed errors range: −11.8% to +7.7%, without a directional trend), versus 27.2% for Weibull + CleanCov. The holdout and full-model March forecasts agree within approximately 3% every week—with $k = 14$ and $T = 41$, this agreement is far from automatic and is consistent with the two-segment architecture capturing genuine behavioral structure rather than merely fitting noise in the training window.

Parsimony: Moderate ($k = 14$); justified by ΔBIC = 6,539 over best single-segment model. Weibull + 2-Segment ($k = 11$) fails: BIC worsens.

Performance: Excellent for Burr XII only. BIC = 4,002,735 (best overall); MAPE = 9.7%—a near-halving vs. CleanCov. Weibull + 2-Segment MAPE = 18.2%, indistinguishable from single-segment.

Robustness: Strong for Burr XII. Holdout MAPE = 5.6%; no systematic drift; March forecasts stable within approximately 3% across model variants. As this is a parameter-heavy model the starting point did affect the parameter estimates.

Generative story: Substantively coherent. Genre-committed fans who buy on recognition vs. mainstream chart followers who need cumulative stock + social legitimation; Weibull shortfall is consistent with Gamma heterogeneity as the structural enabler. *Verdict: Burr XII + 2-Segment is the final model at the expense of parsimony.*

7 Model Comparison

Table 13: Full model progression. BIC $= -2\ell + k\ln(N)$, N = 2,000,000; MAPE over wks 1–41.

Model	Stage	k	Log-Likelihood	BIC	MAPE
Pareto II (No Cov)	No Cov	2	−2,187,695	4,375,419	170.2%
Burr XII (No Cov)	No Cov	3	−2,063,438	4,126,920	87.8%
Exp + Airplay	+Airplay	2	−2,021,612	4,043,254	38.3%
Pareto II + Airplay	+Airplay	3	−2,021,762	4,043,567	38.6%
Burr XII + Airplay	+Airplay	4	−2,015,104	4,030,266	32.2%
Pareto II + FullCov	+FullCov	5	−2,004,632	4,009,337	19.1%
Pareto II + CleanCov	+Clean	4	−2,007,385	4,014,827	24.0%
Burr XII + CleanCov	+Clean	5	−2,005,174	4,010,420	19.4%
Weibull + CleanCov	+Clean	5	−2,004,601	4,009,275	18.0%
Weibull + 2-Segment	+2Seg	11	−2,004,611	4,009,382	18.2%
Burr XII + 2-Segment	**+2Seg**	**14**	**−2,001,266**	**4,002,735**	**9.7%**

The model progression is best understood as a sequence of theoretically motivated structural extensions rather than opportunistic specification search: each tier is justified by specific residual failures documented in the prior tier. The Burr XII + 2-Segment achieves the best performance on all three criteria (BIC, LL, MAPE), with a ΔBIC of 6,539 over the best single-segment model. The Weibull + 2-Segment BIC of 4,009,382 is *worse* than Weibull + CleanCov (4,009,275), suggesting that the two-segment architecture appears to be productive only when the distributional family provides sufficient latent heterogeneity to sustain structurally distinct segment trajectories.

8 March 1996 Forecasts

Table 14: March 1996 forecasts. Weibull + 2-Segment's 18% premium over Burr XII reflects its in-sample over-prediction tendency; the Burr XII's stable ∼31,000/week is the more credible estimate.

Week	Lagged Airplay	Burr XII + 2-Seg	Weibull + 2-Seg
42 (Mar 10)	37,843	32,296	37,349
43 (Mar 17)	37,859	30,412	36,570
44 (Mar 24)	37,679	31,189	35,491
45 (Mar 31)	40,338	31,516	38,972
Total	—	**125,413**	**148,382**

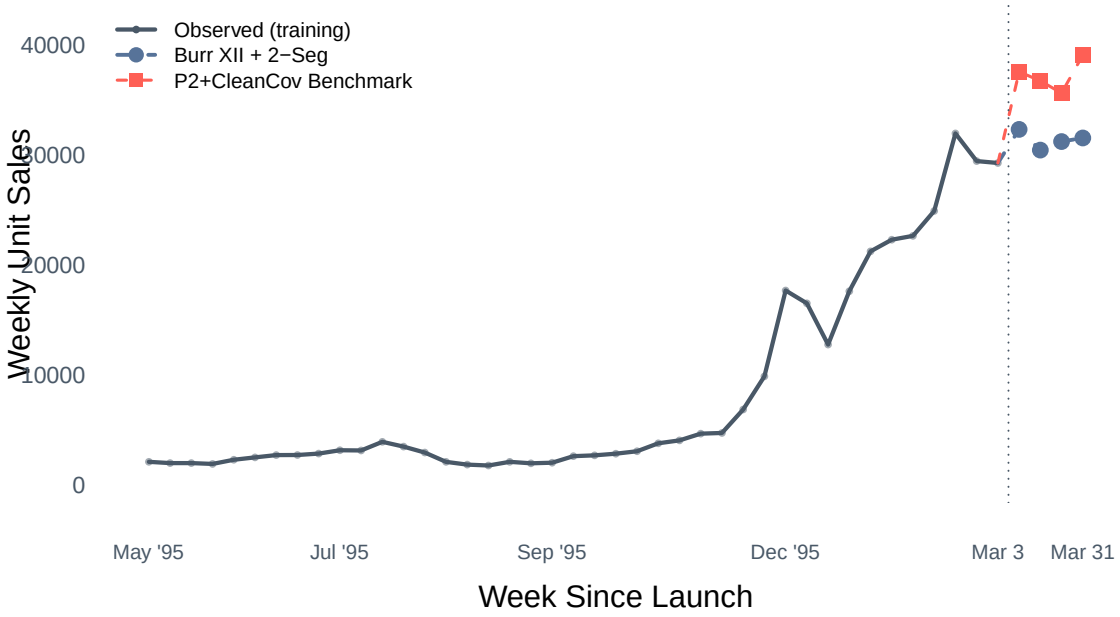


Figure 6: Training sales and four-week March 1996 forecasts. Blue circles: Burr XII + 2-Segment (approx. 31,000–32,000 units/week). Coral squares: Pareto II + CleanCov benchmark (rising to approx. 39,100 by week 45).

The Burr XII + 2-Segment projects approximately 31,000–32,000 units/week through March, totaling approximately **125,413 units**. Week 42 peaks on high inherited lagged airplay and the full breakthrough premium; weeks 43–44 moderate as Segment 2 depletes its remaining survivor pool; week 45 recovers modestly as lagged airplay rises again. By week 41, only 16.97% of the market has adopted, leaving approximately 1.66 million potential adopters untouched. The Pareto II + CleanCov benchmark projects 19% more March units because it has no depletion mechanism—its large survivor pool perpetuates the post-breakthrough uplift indefinitely. The holdout evidence consistently favors controlled depletion over unlimited extrapolation.

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9 Conclusions and Limitations

Main findings. The central result is not merely that two segments fit better—it is that the *specific behavioral signatures* of both segments appear theoretically coherent and mutually

reinforcing. Genre enthusiasts seem to have converted primarily through radio recognition; mainstream buyers through the gradual accumulation of radio exposure and time. The Burr XII's Gamma heterogeneity is arguably not a nuisance parameter in this story: it appears to be the structural element that enables the model to sustain two populations with sharply different hazard trajectories simultaneously. For labels and managers, these results suggest that radio investment and mainstream validation (MTV rotation, chart placement, streaming editorial) may function as complements targeting distinct demand segments, rather than substitutes.

Limitations. *Parameter multicollinearity*: Lagged and cumulative airplay are highly correlated ($r > 0.90$), creating likelihood ridges. The coefficients $\hat{\beta}_{\text{lagAP}}$ and $\hat{\beta}_{\text{cumAP}}$ should be interpreted as the joint radio-exposure effect distributed across timescales, not independent channels; penalized likelihood or principal-component covariates would sharpen individual estimates. This collinearity creates some tension with the segment-specific airplay interpretations presented above: the assignment of lagged airplay sensitivity to Segment 1 and cumulative stock sensitivity to Segment 2 reflects a directional difference that is theoretically coherent and consistent across multiple optimization starting points, but the precise coefficient magnitudes should be treated with appropriate caution rather than taken as stable, independently interpretable channel-level estimates. *Segment-level robustness*: With $k = 14$ and $T = 41$, structural parameters are jointly identified in ways sensitive to starting conditions; forecast-level stability is strong (holdout within approximately 3%), but coefficient-level reliability requires further scrutiny before guiding specific channel-budget decisions. *Fixed market size*: $N = 2{,}000{,}000$ is assumed; sensitivity analysis varying N by $\pm 25\%$ would quantify the dependence of March forecasts on this prior.

Final Model Summary

Burr XII + 2-Segment: $k = 14$, $\ell = -2{,}001{,}266$, BIC $= 4{,}002{,}735$, MAPE $= 9.7\%$, holdout MAPE $= 5.6\%$

Two-segment structure: Seg 1 (8.4%, Genre-Committed Fans): $c = 1.268$, $\hat{\beta}_{\text{lagAP}} = 15.05$, $\hat{\beta}_{\text{brk}} = 3.750$ ($\approx 42\times$); Seg 2 (91.6%, Mainstream Chart Followers): $c = 0.926$, $\hat{\beta}_{\text{cumAP}} = 3.750$ ($\approx 42\times$/SD), $\hat{\beta}_{\text{brk}} = 0.544$ ($\approx 1.7\times$), lagged airplay $= 0$

Key mechanism: Burr XII Gamma heterogeneity enables segment differentiation; Weibull + 2-Segment shows no improvement ($\Delta\ell = -10$, $\Delta k = 6$; both LL and BIC worsen)

March 1996 (wks 42–45): 32,296 | 30,412 | 31,189 | 31,516 $\approx$ **125,413 units**; P2+CleanCov benchmark 19% higher

Market headroom: 16.97% adoption by wk 41 — approx. 1.66 million potential adopters remain

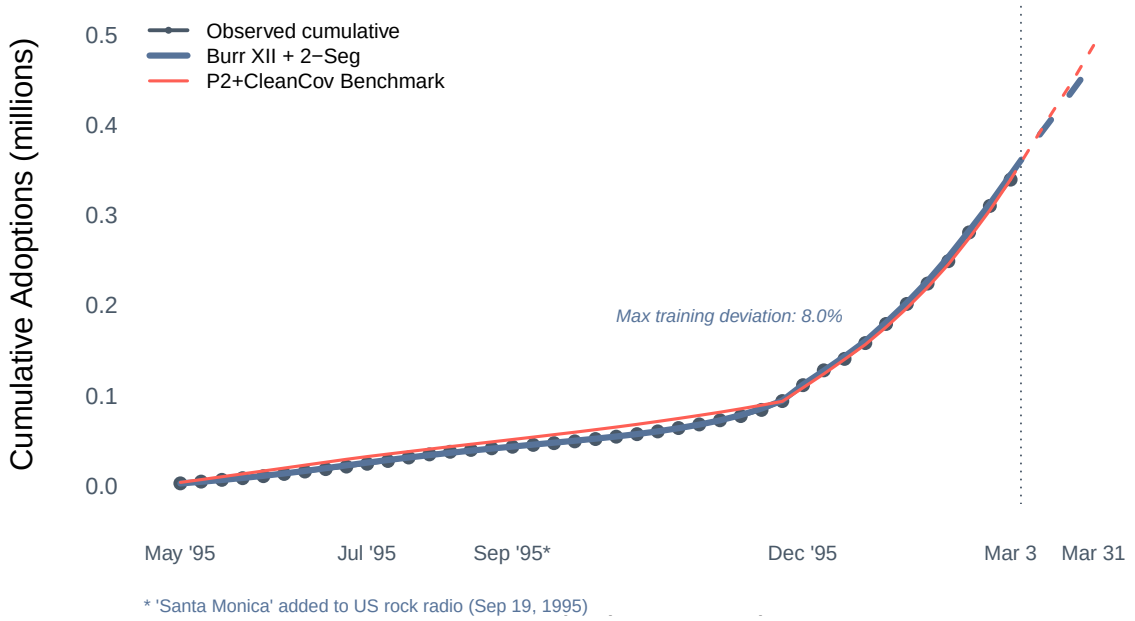


Figure 7: Cumulative observed vs. model-expected adoptions through March 1996. The Burr XII + 2-Segment (blue) tracks observed cumulative sales (black) closely through all 41 training weeks and projects a smooth continuation into the forecast period. The P2+CleanCov benchmark (coral) over-extrapolates post-breakthrough growth, accumulating a 19% excess by week 45. The dashed vertical line marks the end of the training window (week 41).